ALICE L WALTON SCHOOL OF MEDICINE

Supraventricular Arrhythmias: Pathophysiology, Diagnosis, and Treatment

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Adapted from: Supraventricular Arrhythmias

Listen to this Brick
0:00 / 30:11 1x

Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation and pathophysiology of the different types of supraventricular arrhythmias
2. Differentiate [illegible] aventricular arrhythmias.
3. Outline treatment plan for various supraventricular arrhythmias.

CASE CONNECTION

PV, a 64-year-old male, comes to the emergency department with increasing shortness of breath. PV has a history of chronic obstructive pulmonary disease (COPD). "It's been getting worse over the past few days, and I am coughing up more sputum," PV says. "My inhalers aren't helping." On exam, PV is afebrile. His heart rate is 110 beats/min and irregular, blood pressure is 142/78 mm Hg, and respiratory rate is 26 breaths/min. He has decreased breath sounds but no rales. Treatment for his COPD flare begins, and you turn your attention to his cardiac status. His ECG shows a narrow QRS complex irregular rhythm with a rate of 112/min. You see a few different P waves.

What is PV's rhythm disturbance? How should PV's arrhythmia be treated? Consider your answers as you read, and we'll revisit PV at the end of the brick.

GO TO CONCLUSION ↓

What Are Supraventricular Arrhythmias?

Supraventricular arrhythmias are abnormal heart rhythms that originate above the ventricles, most commonly in the atria. QRS complexes are usually narrow (normal) because the impulses are conducted along the normal pathway involving the AV junct[illegible]tion occurs with bundle branch diseas[illegible]

Supraventricular arrhythmias are typically less serious than their ventricular counterparts. They are less likely to cause cardiac arrest compared to ventricular arrhythmias.

Types of Supraventricular Arrhythmias (LO#1–3)

The types of arrhythmias we cover here are premature atrial contractions, atrial flutter, atrial tachycardia, multifocal atrial tachycardia, atrioventricular nodal reentrant tachycardia (AVNRT), and atrioventricular reentrant tachycardia (AVRT). Atrial fibrillation, another supraventricular arrhythmia, has its own separate brick and is not covered here.

Usage note on the term "supraventricular tachycardia"

Although all the above arrhythmias are supraventricular, and most present with tachycardia, the term "supraventricular tachycardia" (or, as many publications name it, "paroxysmal supraventricular tachycardia" or PSVT) is more restrictive in common usage. It is defined as a regular narrow (usually) complex tachycardia of abrupt onset and termination with a rate range of 110 to 200 bpm. Although the term has this official definition, it is a *category*, an umbrella term rather than a specific arrhythmia. It subsumes other specific arrhythmias, namely AVNRT, AVRT and some types of atrial tachycardia, all of which will be discussed below. It is important to note that the term PSVT does NOT include atrial fibrillation or flutter.

Sinus tachycardia in relation to the supraventricular arrhythmias

Sinus tachycardia, th[illegible]iologic increase in the sinus rate in respo[illegible] is thus not considered an arrhythmia. Though not covered in detail here it deserves mention because, depending on the clinical context, it may be more serious than the supraventricular arrhythmias themselves, as it can be a warning of serious illness, even a life threatening situation (eg heart failure, pulmonary embolism, sepsis). It is also worth mentioning that at higher rates, sinus tachycardia can be confused with AVNRT, AVRT and atrial flutter with 2:1 conduction ratios. Distinguishing features pointing to sinus tachycardia include normal P waves (which may be difficult to see at very fast rates), and gradual onset and resolution (in contrast to the supraventricular arrhythmias discussed here which have abrupt onset and termination).

What Are Premature Atrial Contractions? (LO#1and 3)

PACs are defined as abnormal discharges from the atria which arise from an area outside the sinoatrial node (and are thus considered "ectopic") and occur prematurely, that is, before completion of the cardiac cycle. They may be experienced by patients as an extra heartbeat or "skipping a beat." They are the least dangerous of these arrhythmias.

Clinical Presentation

PACs can be asymptomatic, or patients may complain of a skipped beat.

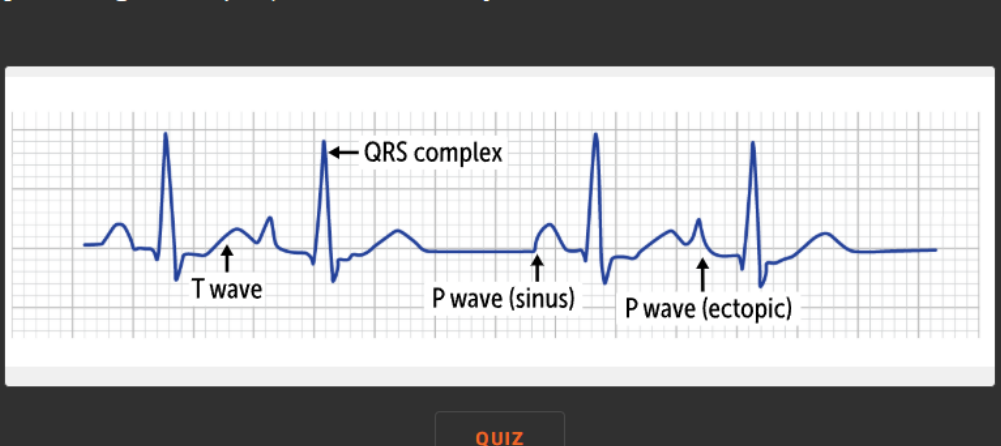
Pathophysiology

They arise from an ectopic focus within the atria, producing an abnormal P wave on ECG. They can occur in almost any type of structural heart disease or in patients with no k[illegible]

Secondary (external) causes. PACs may be worsened by stress or stimulant drugs, although often they occur for no apparent reason.

Diagnosis

On ECG, PACs produce aberrant P waves that differ from typical P waves. They appear early in the cardiac cycle (at a shorter time interval as compared to the prevailing sinus cycle) hence the term *premature*.



QUIZ

Figure 1 Premature atrial contraction

These P waves are usually *conducted* (that is, just as in sinus rhythm, the ectopic P wave is followed by a QRS complex). Occasionally, if the ectopic P wave occurs so early after the preceding beat that the AV junction has not had time to recover from its refractory period, the PAC will be *nonconducted* (not followed by a QRS).

Management

PACs generally require no treatment. Depending on the clinical circumstance, they may warrant additional testing for underlying structural heart disease.

Why would you need to further test the otherwise healthy patient if they develop frequent PACs?

What Is Atrial Flutter? (LO#1 and 3)

Atrial flutter is a type of supraventricular tachyarrhythmia characterized by rapid but organized electrical activity in the atria. This condition often presents as a racing heart rate and can affect how effectively the heart pumps blood.

Clinical Presentation

Atrial flutter presents [illegible] other tachyarrhythmia can cause palpitations, symptoms of low cardiac output (eg dizziness), exacerbation of existing heart disease such as heart failure of ischemia (hence causing dyspnea or chest pain).

Pathophysiology

In typical atrial flutter, instead of normal electrical impulses originating from the sinoatrial (SA) node, a macro-reentrant circuit develops in the right atrium. This circuit causes the atria to contract at a rapid rate, typically between 240 to 340 beats per minute. In typical atrial flutter the waves (sometimes referred to as F waves) cause "Sawtooth Pattern" mainly in leads II, III, and aVF. Because the whole atrium is involved in the macro-reentrant circuit ... and most of us have the same size atrium ... the circuit typically conducts at 300 beats per minute. The AV Node, because of its inherently slow conduction and filtering function, will not conduct to the ventricles that fast, so atrial flutter typically conducts at the next slower integer of 300 bpm which is 150 bpm (2:1 conduction). Ratios show atrial cycles to ventricular cycles. This heart rate at 150 bpm causes a sensation of rapid palpitations for most patients. This arrhythmia can also conduct at other integers of 300 bpm (150 = 2:1, 100 = 3:1, 75 = 4:1, 60 = 5:1, etc).

Atrial flutter can have atypical forms in which the macroreentry is at other locations such as the mitral annulus (perimitral flutter), left atrial roof, and regions of atrial scarring in either the right or left atrium. In those cases, organized depolarizations of the atria at a rate of around 300 are still seen, but they may not exhibit the typical saw tooth pattern.

Underlying Causes. Almost any acute or chronic structural heart disease can cause atrial flutter.

Diagnosis

On ECG, *typical* AFL has a P wave **"sawtooth" pattern** most distinctly seen in leads II, III, and aVF (Figure 3).

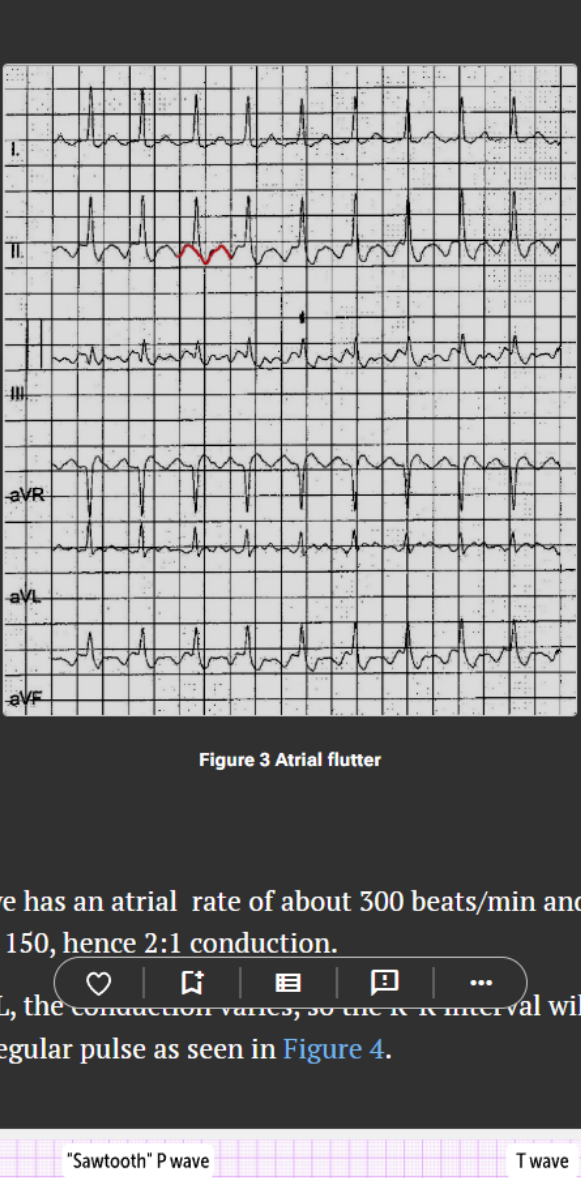
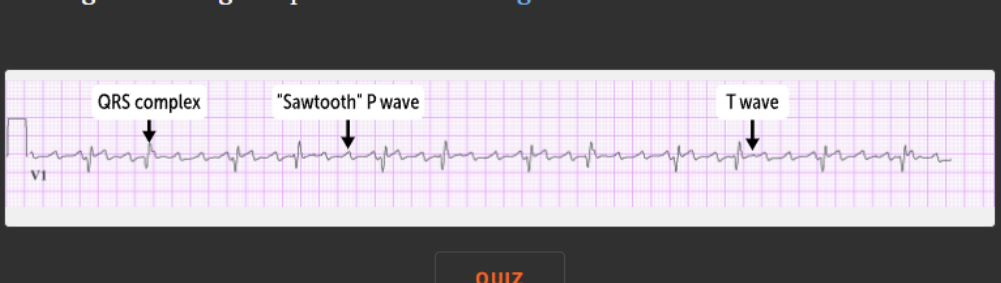


Figure 3 Atrial flutter

The tracing above has an atrial rate of about 300 beats/min and a ventricular rate of half that, 150, hence 2:1 conduction.

Less often in AFL, the [illegible]val will then vary, leading to an irregular pulse as seen in Figure 4.



QUIZ

Figure 4 Atrial flutter with variable conduction

To summarize, the atrial impulses (flutter waves), because they are formed by a constant velocity reentry circuit, have a constant rate. Variation in the ventricular rate occurs solely due to changes in conduction rations. Put another way, ventricular rate variations in atrial flutter occur as *multiples*. This creates the potential for *abrupt*, rather than *gradual*, changes in the ventricular rate, making atrial flutter a potentially more unstable rhythm than atrial fibrillation.

Management

Acute Management: Rate control with beta-blockers or calcium channel blockers, is the first line of therapy. Rhythm control (converting to, and maintaining, sinus rhythm), employs antiarrhythmic medications or cardioversion if necessary. Long-term Management: Ongoing anticoagulation is often recommended to prevent stroke, especially in patients with risk factors. In some cases, cathet[illegible]inate the abnormal electrical circuit. (It is important to note that atrial flutter carries a stroke risk that is similar, though slightly lower in some reports, to atrial fibrillation. Current guidelines indicate that atrial flutter and fibrillation should be treated the same when it comes to stroke prevention with anticoagulation).

Atrial tachycardia

Atrial tachycardia (aka focal atrial tachycardia) is similar to atrial flutter in that it is characterized by abnormally rapid, regular atrial depolarizations. There are essential differences from flutter in terms of rate, the appearance of the abnormal atrial depolarizations, and the electrophysiologic mechanisms.

The official definition, according to the American College of Cardiology/American Heart Association/Heart Rhythm Society, is that it is a regular atrial rhythm at a constant atrial rate of >100 bpm with discrete P waves and atrial activation sequences originating outside of the sinus node. It differs from atrial flutter in several respects. First, the rate is generally slower and because of this, the AV node can usually handle 1:1 conduction. The usual conduction ratio is 1:1 although occasionally 2:1 or even higher ratios may be seen. Unlike atrial flutter, which is always due to macroreentry, the electrophysiologic mechanisms of atrial tachycardia are diverse and include enhanced automaticity, triggered activity, and micro (not macro) reentry.

Atrial tachycardia, like some other supraventricular arrhythmias, has diverse causes. One particularly noteworthy cause is digitalis intoxication. Digitalis medications, such as digoxin, enhance the effects of the vagus nerve in the heart, so that when atrial tachycardia is caused by digitalis preparations 2:1 or higher conduction rations are seen, which are unusual in other causes of atrial tachycardia. Thus, if atrial tachycardia with conduction rations other than 1:1 is seen in a patient ta[illegible]ately come to mind.

What Is Multifocal Atrial Tachycardia? (LO#1and3)

MAT is an irregular tachycardia (>100 beats/min) seen most often in patients with lung disease.

Clinical Presentation

MAT is often asymptomatic, and the pulmonary symptoms of the lung disease are dominant. It can sometime show the same rate-related symptoms described above under AF or AFL.

Electrocardiographic pattern

MAT occurs when **three or more distinct impulses** arise in the atria; all travel to the AV node and cause depolarization. It presents as an irregular tachyarrhythmia, like atrial fibrillation. Unlike atrial fibrillation, however, it has P waves. These occur irregularly at a fast rate and have different morphologies (at least 3 according to the definition). In a sense, it resembles sinus tachycardia with multiform PACs.

What ch[illegible]iogram?

Causes.

MAT occurs mostly in the setting pulmonary disease, particularly during exacerbations (eg, COPD, pneumonia). Other causes include cardiac diseases, hypokalemia, hypomagnesemia, CKD, and digitalis toxicity.

Diagnosis

On ECG, MAT shows an irregular tachycardic rhythm, because the timing of depolarization differs from site to site, and at least three different morphologies of P waves (Figure 5).

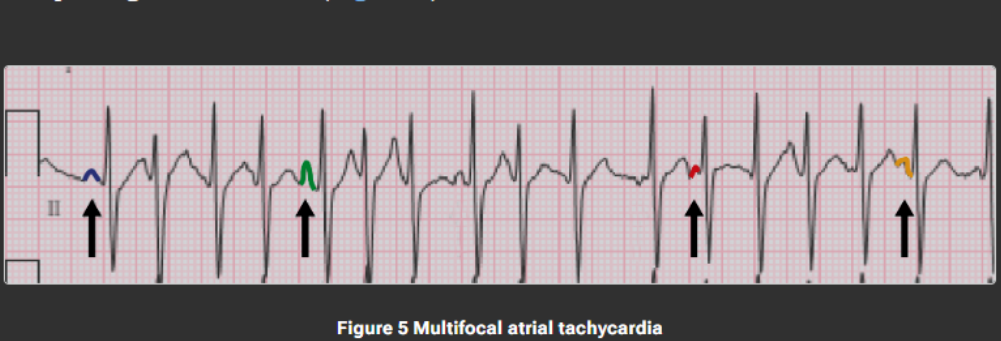


Figure 5 Multifocal atrial tachycardia

Atrial and ventricular [illegible]min.

Management

In most cases, treating the underlying pulmonary disease or treatable secondary causes will restore normal sinus rhythm. Optimization of pulmonary function and correction of electrolyte disturbances are important. In some cases, β-blockers or nondihydroipyridine calcium channel blockers (CCBs) like diltiazem can be used.

What Is Atrioventricular Nodal Reentrant Tachycardia? (LO#1and3)

AVNRT is a subset of PSVT as described above. It is an abruptly occurring (and terminating) regular narrow complex tachycardia with a rate range of 110-220 bpm. There may be no discernible p waves, or one may see inverted (retrograde) atrial impulses. It is officially defined by its mechanism: micro reentry in the AV node in a circuit created by dual AV nodal pathways (a fast and a slow pathway). *(AVNRT is a micro-reentrant tachycardia involving dual AV nodal pathways (fast and slow pathways) with participation of perinodal atrial tissue, with the reentrant circuit functionally centered on the AV node.)*(previously discussed pathophysiology brick)

Clinical Presentation

Patients have the same rate-related symptoms as with other tachyarrhythmias. Because it often occurs in otherwise healthy patients with no structural heart disease, symptoms may be mild and it is usually not life threatening.

Underlying cause

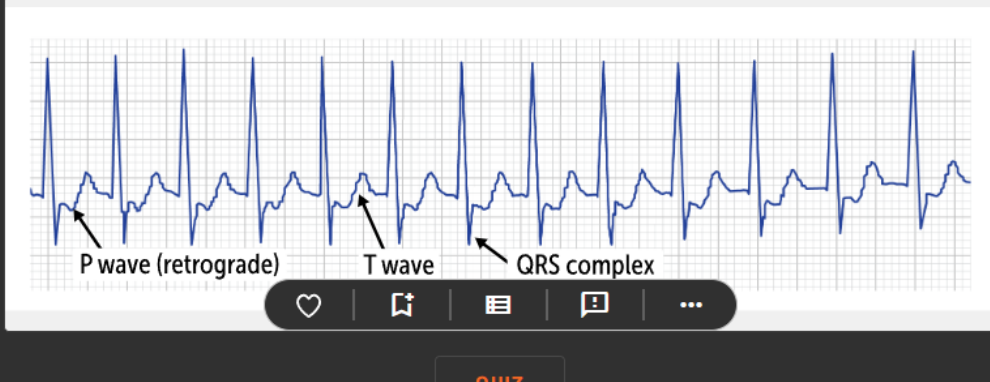
Many patients with AVNRT have no underlying cardiac pathology. The dual AV nodal pathways that form the substrate for this arrhythmia are not considered pathological, but rather an anatomic variation. This variation is common, existing in 7-10% of the population, with a prevalence roughly equal to that of being left handed. This is the MOST COMMON SVT in young people.

Triggers

AVNRT is triggered by a PAC, so the causes of PACs discussed above can be triggers for AVNRT. In some cases, no trigger can be identified.

Diagnosis

On ECG, AVNRT shows a sudden onset of rapid, narrow QRS complexes at a rate of 120 to 220 beats/min. These narrow QRS complexes are each followed by a "retrograde" P wave, which represents the impulse being conducted backward from the AV node to depolarize the atrium (Figure 7). The retrograde P wave, however, is not always seen, and may be "buried" in the QRS. If it is seen, it's usually seen in lead V1.



QUIZ

Figure 7 Atrioventricular nodal reentry tachycardia

Management

AVNRT can be treated with several maneuvers. For stable patients, treatment options include vagal maneuvers and/or medications. Vagal maneuvers such as Valsalva (forced exhalation against a closed glottis) and carotid sinus massage increase vagus nerve tone and therefore slow the AV nodal transmission rates; they are effective in 25% of cases. The medication adenosine is effective in more than 90% of cases and is frequently used. Alternative medications such as β-blockers and verapamil, a calcium channel blocker, may also be useful.

CLINICAL CORRELATION

If the patient is unstable or does not respond to the above, instead use synchronized cardioversion. Synchronized cardioversion delivers a timed electrical shock that is synchronized with the R wave of the ECG to terminate unstable tachyarrhythmias. This timing avoids delivery during the vulnerable repolarization phase, reducing the risk of inducing ventricular fibrillation.

What Is Atrioventricular Reentrant Tachycardia? (LOs#1 and 3)

AVRT is a manifestation of Wolff Parkinson White syndrome (WPW) in which a congenital aberrant conduction pathway (Kent bundle) connects the atria and ventricles, bypassing the AV node. Clinically, and on the electrocardiogram, AVRT superficially resembles AVNRT and can be difficult to distinguish from it. It differs electrophysiologically however in that, unlike AVNRT, which is driven by *micro* reentry, AVRT is caused by *macro* reentry, in a circuit through which the reentrant impulse travels retrogradely through the aberrant pathway and antegradely down the AV junction and bundle branches, thus resulting in a narrow and normal appearing QRS.

Diagnosis

During tachycardic episodes the ECG closely resembles AVNRT and may be difficult to distinguish from it. Due to the larger size of the reentry circuit (macro instead of micro as is the case with AVNRT) the retrograde atrial impulses occur slightly later in the cardiac cycle and are not up against the QRS, making them more likely to be seen.

The baseline ECG (during sinus rhythm) is also abnormal in WPW. The Clinical Triad includes a shortened PR interval (<120 msec), a characteristic "delta wave" (slurred upstroke of the R wave) and QTc prolongation (Figure 8, Figure 9). Accessory pathway conduction is slower than the His-Purkinje system. The delta wave is due to initial ventricular depolarization originating from the accessory pathway inserting directly into the ventricular muscle. This phenomenon is known as pre-excitation. The size of the delta wave is proportional to the volume of ventricular muscle that depolarizes from the bypass tract. Remember, the QRS complex is a "summation" of electrical depolarization of the [illegible]D fast conduction.

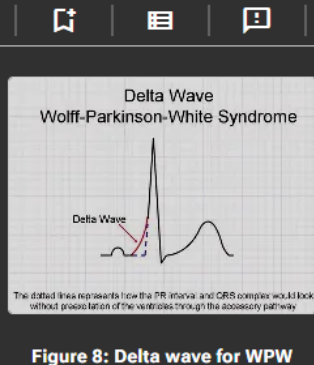


Figure 8: Delta wave for WPW

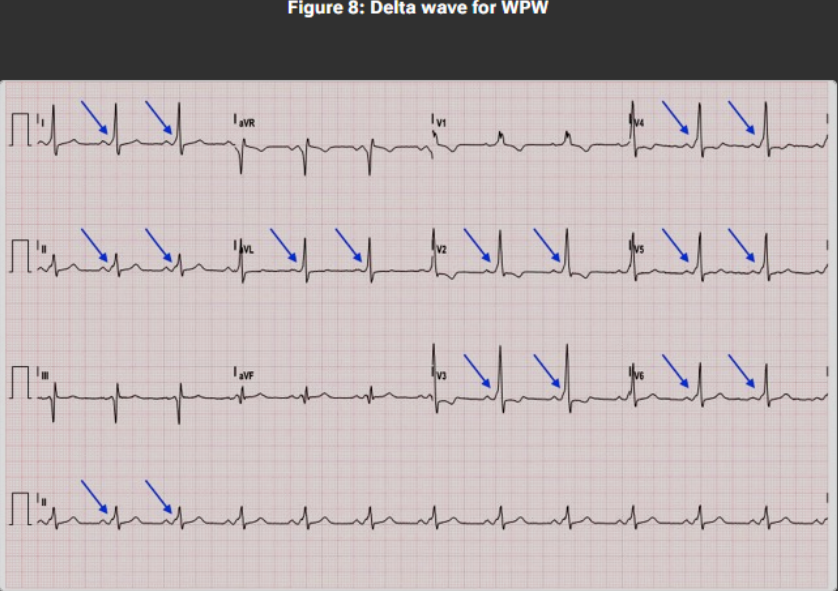


Figure 9: WPW with Delta wave

Besides standard ECG, stress testing and electrophysiologic studies are often done to reveal the source of the arrhythmia.

Management

For a stable patient who is acutely tachycardic and symptomatic, consider adenosine, β-blockers, and non-dihydropyridine calcium blockers, in much the same way that AVNRT is treated. A note of caution: all these agents are contraindicated in AVRT/WPW if the patient goes into atrial fibrillation. The explanation is beyond the scope of this brick.

CLINICAL CORRELATION

Once the patient is not acutely tachycardic, the first-line treatment is catheter radioablation of abnormal pathways, done in the electrophysiology lab.

Why do you note a characteristic delta wave prior to each QRS complex in a patient with Wolff-Parkinson-White (WPW) syndrome?

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to PV, what is his rhythm disturbance? How should his arrhythmia be treated?

The ECG highlights four important points that lead to the diagnosis of PV's rhythm disturbance: increased rate (tachycardia), narrow QRS complex, irregular rhythm, and three or more distinctly different P waves. PV is in multifocal atrial tachycardia, which is secondary to his underlying COPD. After treating his COPD flare, his MAT resolves, restoring normal sinus rhythm. When you look at an ECG done 24 hours after admission, PV is in normal sinus rhythm.

Summary

- Supraventricular arrhythmias are deviations from normal sinus rhythm arising above the ventricles.
- Supraventricular arrhythmias are less likely to cause cardiac arrest compared to ventricular arrhythmias.
- Premature atrial contractions (PACs) are common, minimally symptomatic arrhythmias that cause an extra heartbeat.
- PACs are usually harmless and require no treatment.
- Atrial fibrillation (AF) is a supraventricular tachycardia in which the atria fibrillate, or quiver. (Covered in a separate brick).
- Atrial flutter (AFL) is a supraventricular tachycardia in which the atria beat faster than the ventricles.
- The ECG baseline with *typical* AFL is described as a sawtooth pattern, most distinct in leads II, III, and avF.
- Multifocal atrial [illegible]with chronic lung disease, especially chronic obstructive pulmonary disease (COPD).
- MAT shows multiple foci of P waves on ECG, with at least three different morphologies.
- Atrioventricular nodal reentrant tachycardia (AVNRT) occurs because of the presence of slow and fast pathways in the AV node that conduct at different rates, allowing a reentry circuit to set up inside the AV node.
- ECG findings in AVNRT include narrow QRS complexes with retrograde P waves sometimes seen, and a heart rate more than 100 beats/min (typically 150-200 bpm).
- Stable patients with AVNRT may be treated with vagal maneuvers (Valsalva or carotid sinus massage), medications like adenosine, or alternatives like β-blockers and verapamil.
- Atrioventricular reentrant tachycardia (AVRT) is a tachyarrhythmia due to abnormal AV pathways, due to Wolff-Parkinson-White syndrome. The QRS complex is generally narrow and normal appearing during the tachycardia. After conversion to sinus rhythm one typically sees the abnormal QRS morphology (delta wave and distorted QRS) associated with WPW.
- Potentially curative treatment of WPW may involve radioablation of the abnormal pathway (Kent bundle).

Review Questions

1. **Which type of arrhythmia most likely shows a delta wave during sinus rhythm?**

A. ○ Atrial fibrillation
B. ○ Atrial flutter
C. ○ Atrioventricular nodal reentrant tachycardia
D. ○ Premature atrial complexes
E. ● Wolff-Parkinson-White (WPW) syndrome

Hide Explanation

The correct answer is Wolff-Parkinson-White (WPW) syndrome (E) because WPW is characterized by an extra electrical pathway that causes ventricular preexcitation. This translates into a delta wave on ECG. None of the other arrhythmias in the choices display delta waves. Delta waves are unique to WPW.

2. **A female is brought to the emergency department after she felt her heart beat very fast and nearly passed out. An ECG shows typical atrial flutter with 2:1 conduction, with a ventricular rate of 150 beats/min. Which of the following ECG findings is most specific for this type of arrhythmia?**

A. ○ Delta wave
B. ○ Indistinguishable P waves
C. ○ Retrograde P waves
D. ● Sawtooth[illegible]
E. ○ Wide QRS complexes

Hide Explanation

The correct answer is sawtooth pattern (D), which are inverted but rapid P waves. A delta wave (A) is a characteristic finding in Wolff-Parkinson-White syndrome, which occurs due to conduction along the aberrant fast pathway. Indistinguishable P waves (B) are a finding of atrial fibrillation. Retrograde P waves (C) are found in atrioventricular nodal reentrant tachycardia due to the impulse being conducted backward from the AV node to depolarize the atrium. Wide QRS complexes (E) are seen most often with ventricular, not supraventricular arrhythmias like atrial flutter.

3. **Which of the following management strategies is least likely to be helpful in a patient with atrioventricular (AV) nodal reentrant tachycardia?**

A. ○ Adenosine
B. ● Amlodipine
C. ○ β-blockers
D. ○ Carotid sinus massage

Hide Explanation

The correct answer is [illegible] blocker, is a dihydropyridine calcium blocker. Dihydropyridine calcium channel blockers, at standard doses, have little or no effect on the AV node, which is where the reentry circuit is taking place. Nondihydropyridine calcium blockers such as verapamil and diltiazem may be effective in AVNRT but they were not among the choices listed. All of the other treatment choices listed can be effective in terminating AVNRT.

4. **Which of the following supraventricular tachyarrhythmias is most likely to be terminated using the Valsalva maneuver?**

A. ○ Atrial fibrillation
B. ○ Atrial flutter
C. ● Atrioventricular nodal reentrant tachycardia
D. ○ Multifocal atrial tachycardia

Hide Explanation

The correct answer is atrioventricular nodal reentrant tachycardia (C). Vagal maneuvers like Valsalva or carotid sinus massage increase vagal tone and slow the atrioventricular (AV) nodal conduction rate, effectively obliterating the tachyarrhythmia in up to 25% of cases. The sustaining mechanism for AVNRT depends on a narrow and critical range of conduction velocity in the AV node. If conduction is slowed outside this range, either by vagal maneuvers or certain drugs, the arrhythmia cannot [illegible] atrial fibrillation and atrial flutter (choices [illegible]ease conduction to the ventricles and thereby slow the ventricular rate. However, this will not terminate those arrhythmias, because their sustaining mechanisms lie not in the AV node but in the atria. Multifocal atrial tachycardia (D) also originates in the atria, such that vagal input, which works on the AV node, will not stop the sustaining mechanism.

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the clinical presentation and pathophysiology of the different types of supraventricular arrhythmias~~
02 ☑ ~~Differentiate the ECG patterns associated with supraventricular arrhythmias.~~
03 ☑ ~~Outline treatment plan for various supraventricular arrhythmias.~~

Objective Confidence: 3/3